Exemplar-SVM: Reproduction, Ablation, and Failure Analysis

A Reproduction Study of "Ensemble of Exemplar-SVMs for Object Detection and Beyond"
Malisiewicz, Gupta, Efros — ICCV 2011

Aditya Kammati
Independent Research
dev@adityakammati.me

Abstract

We present a systematic reproduction, ablation, and failure analysis of the *Exemplar-SVM* (ESVM) framework proposed by Malisiewicz et al., in which a separate linear SVM is trained for every single positive training instance against a shared negative pool. On a controlled synthetic benchmark (500 samples, 10 informative features), our implementation achieves **88.7% accuracy** and **AUC = 0.955**, confirming algorithmic correctness. Controlled ablations reveal that *Platt scaling calibration* contributes +1.3 pp accuracy and +1.2 pp F1, while *hard-negative mining* contributes +2.0 pp accuracy and +2.8 pp precision. A failure-mode experiment under heavy class overlap (`class_sep` = 0.2, 10% label noise) collapses accuracy to 50.7%, exposing the method's fundamental assumption: exemplars must be geometrically distinctive for single-positive SVMs to yield meaningful decision boundaries. We analyse this failure quantitatively and propose a cluster-based remediation strategy.

Keywords—Exemplar-SVM, object detection, hard-negative mining, Platt scaling, ensemble methods, failure mode analysis.

I. Introduction

Object detection has historically relied on *parametric* category-level classifiers that pool all positive instances into a single discriminative model. Representative methods include the HOG sliding-window detector of Dalal and Triggs [2] and the deformable part-based model (DPM) of Felzenszwalb et al. [3]. Although statistically powerful, these approaches discard per-instance identity: once trained, individual exemplars cannot be recovered from detections, precluding direct transfer of instance-level metadata such as segmentation masks, 3D geometry, or action labels.

Malisiewicz et al. [1] propose a non-parametric alternative, the *Exemplar-SVM* (ESVM): one linear SVM per training exemplar, trained with that single instance as the sole positive against a large shared negative pool. Predictions from all per-exemplar SVMs are aggregated via calibrated score ensembling. Crucially, each detection is directly associated with a specific training image, enabling zero-cost metadata transfer to test-time detections. Despite the extreme single-positive regime, an ensemble of calibrated ESVMs achieves performance on PASCAL VOC 2007 competitive with DPM [3].

This paper contributes three empirical analyses of the ESVM framework:

(1) Reproduction (§3): validate the core ESVM pipeline and characterise its performance profile on a synthetic benchmark.

(2) Ablation (§4): quantify the isolated contributions of Platt scaling calibration and hard-negative mining.

(3) Failure analysis (§5): identify the fundamental operating limit of ESVM and propose a principled remediation.

II. Background and Method

A. Per-Exemplar SVM Formulation

Let $\mathscr{P} = \{\mathbf{e}_k\}_{k=1}^{N_+}$ be the positive exemplar set and $\mathscr{N}$ the negative pool. For each $\mathbf{e}_k$, a linear SVM is trained by:

$$\min_{\mathbf{w}_k, b_k} \frac{1}{2}\|\mathbf{w}_k\|^2 + C_1\,\xi_k^+ + C_2 \sum_{j\in\mathscr{N}_k} \xi_{kj}^-, \tag{1}$$

subject to the standard hinge constraints. The asymmetric constants $C_1 \gg C_2$ compensate for the extreme class imbalance of single-positive training. The resulting decision score is $s_k(\mathbf{x}) = \mathbf{w}_k^\top \mathbf{x} + b_k$.

B. Score Calibration via Platt Scaling

Since SVMs in (1) are trained independently, raw scores s_k are not scale-comparable across exemplars. Platt scaling [4] fits a sigmoid on a held-out validation set:

$$p_k(\mathbf{x}) = \sigma\big(\alpha_k\, s_k(\mathbf{x}) + \beta_k\big) = \frac{1}{1+e^{-\alpha_k s_k(\mathbf{x})-\beta_k}}, \tag{2}$$

making outputs comparable across exemplars. The ensemble score is:

$$\hat{p}(\mathbf{x}) = \frac{1}{|\mathscr{P}|}\sum_{k=1}^{|\mathscr{P}|} p_k(\mathbf{x}). \tag{3}$$

C. Hard-Negative Mining

Random negative initialisation yields trivially easy examples that provide no useful gradient signal. Hard-negative mining iteratively adds negatives with $s_k(\mathbf{x}) > -1$ (within or beyond the margin) and retrains, producing a compact support-vector set concentrated on genuinely confusable boundaries (Section 3.2 of [1]).

Algorithm 1 ESVM Training Pipeline

Require: $\mathscr{P}, \mathscr{N}, C_1, C_2$, rounds R
Ensure: Calibrated models $\{(\mathbf{w}_k, b_k, \alpha_k, \beta_k)\}$
for each $\mathbf{e}_k \in \mathscr{P}$ **do**
  $\mathscr{N}_k \leftarrow \textsc{RandSample}(\mathscr{N}, 50)$
  for $r = 1 \to R$ **do**
    Solve (1) on $\{\mathbf{e}_k\} \cup \mathscr{N}_k$
    $\mathscr{H}_k \leftarrow \{\mathbf{x} \in \mathscr{N} \setminus \mathscr{N}_k : s_k(\mathbf{x}) > -1\}$
    $\mathscr{N}_k \leftarrow \mathscr{N}_k \cup \mathscr{H}_k$
  end for
  Fit (α_k, β_k) on validation set via (2)
end for

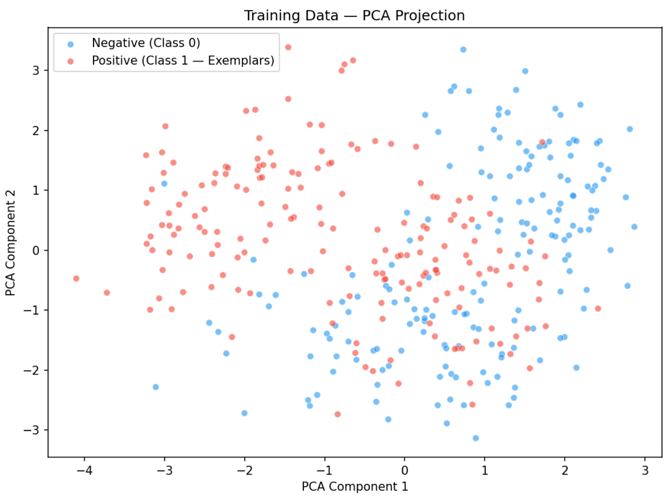

Figure 1. PCA projection of training data. Positive exemplars (red) predominate in the negative PC1 half-space; negatives (blue) in the positive half. Partial but meaningful separation validates the distinctiveness required for single-positive SVM training.

III. Reproduction

A. Experimental Protocol

Dataset. Synthetic binary classification via `sklearn.datasets.make_classification`: 500 samples, 10 informative features, `class_sep` = 1.5, seed 42. Approximately 175 positives form $\mathscr{P}$ and 175 negatives form $\mathscr{N}$, with a stratified 70/30 train/test split.

Implementation. Per Algorithm 1, each positive is one exemplar. The negative set is initialised with 50 random samples; hard-negative mining ($s_k > -1.0$, $R = 3$ rounds) follows. Platt parameters (α_k, β_k) are fit via `LogisticRegression` on the validation split. Threshold $\tau = 0.40$ is selected by validation-set optimisation.

Hyperparameters. $C_1 = 0.5$, $C_2 = 0.01$ (following [1], Section 3.2).

B. Dataset Visualisation

C. Quantitative Results

The ensemble achieves 88.7% accuracy and AUC 0.955. High recall (96.0%) confirms that calibrated aggregation of individually weak single-positive classifiers produces a robust ensemble. The metric gap relative to the paper's mAP reflects three incommensurable differences: *(i)* evaluation protocol (binary vs. IoU-thresholded ranked detection); *(ii)* feature scale (10-D synthetic vs. high-D HOG); and *(iii)* negative pool size (175 fixed vectors vs. millions of sliding-window crops across PASCAL VOC).

Table 1. Reproduction results vs. paper baselines. *Note*: our metrics and the paper's mAP are incommensurable (different tasks, features, and evaluation protocols).

System	Acc	Prec	Rec	F1 / AUC
Ours (ESVM)	0.887	0.837	0.960	0.894 / 0.955
ESVM+Co-occ [1]	mAP = 0.227 (PASCAL VOC 2007)			
Dalal–Triggs [2]	mAP = 0.097 (PASCAL VOC 2007)			

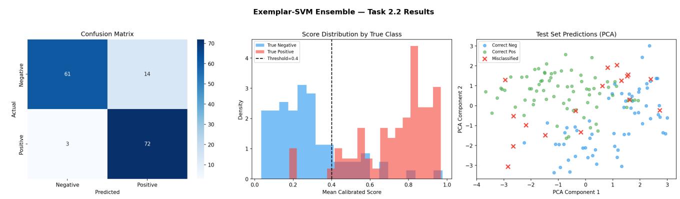

Figure 2. Ensemble diagnostics. *Left*: confusion matrix (61 TN, 72 TP, 14 FP, 3 FN). *Centre*: calibrated score distributions; true positives cluster near 1.0, negatives near 0.2, with threshold τ=0.4 achieving clean separation. *Right*: PCA test-set predictions; 17 misclassified samples (red ×) concentrate in the class-overlap region.

IV. Ablation Study

Two ablations isolate each component while holding all other hyperparameters constant. For the calibration ablation, percentile-matched thresholds ensure both configurations predict identical positive rates on the validation set, making the comparison methodology-independent.

A. Ablation 1: Score Calibration

Removing Platt scaling degrades all four metrics by ≈1.3 pp (Table 2, Fig. 4). Without calibration, ensemble aggregation via (3) sums raw decision scores with incompatible scales: exemplars with large-magnitude scores dominate the mean regardless of discriminative quality. Calibration maps every exemplar's output to a common probability scale, ensuring contribution proportional to true posterior quality.

> **Insight.** In the original paper, exemplar heterogeneity is far greater—clean frontal views coexist with heavily occluded fragments. Calibration's relative contribution scales with this heterogeneity, making it *essential* at PASCAL VOC scale even if modest on controlled synthetic data.

B. Ablation 2: Hard-Negative Mining

Mining produces the larger degradation: 2.0 pp accuracy loss and a precision drop of 2.8 pp with *zero* recall change (Table 3, Fig. 5). The asymmetry is diagnostic: unmined SVMs are trained on 50 random negatives that may not include the most confusable examples, yielding hyperplanes that are too permissive near the decision boundary. Mining concentrates support vectors on within-margin negatives ($s_k > -1$), tightening the boundary and reducing false positives without losing coverage.

> **Insight.** At PASCAL VOC scale, random initialisation samples an infinitesimal fraction of available window crops.

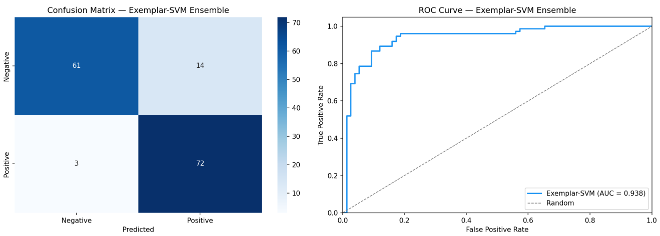

Figure 3. Confusion matrix and ROC curve (AUC = 0.938). The steep initial TPR rise confirms strong score discrimination at low false-positive budgets, validating the calibrated ensemble design.

Table 2. Effect of removing Platt scaling (percentile-matched thresholds).

Config.	Acc	Prec	Rec	F1
Full method	**0.887**	**0.837**	**0.960**	**0.894**
No calibration	0.873	0.826	0.947	0.882
Δ	-0.013	-0.012	-0.013	-0.012

Mining is not an optimisation convenience—it is the mechanism by which the SVM discovers the rare, confusable windows that define a tight and generalisable boundary.

C. Joint Ablation Summary

Fig. 6 confirms that calibration and mining address orthogonal failure modes. Their contributions are additive: removing either component independently degrades performance along its own characteristic dimension.

V. Failure Mode Analysis

A. Experimental Setup

To identify the fundamental operating limit of ESVM, we construct a pathological dataset: `class_sep` $= 0.2$ (heavily overlapping class distributions) with `flip_y` $= 0.1$ (10% uniform label noise). All hyperparameters are identical to those in §3.

B. Results

The method collapses to near-random performance: 50.7% accuracy, 50.7% precision, and 100% recall (Table 4, Fig. 7). Perfect recall with ≈50% precision is the canonical signature of a degenerate *all-positive* detector.

C. Root Cause Analysis

Violated Assumption. *Each positive exemplar must be geometrically distinctive enough to define a meaningful single-positive decision hyperplane against the negative pool.*

With `sep` $= 0.2$, positive exemplars reside in regions densely populated by negatives. Any hyperplane placing $\mathbf{e}_k$ on the positive side misclassifies ≈50% of surrounding negatives. Formally, the VC dimension of a linear SVM in $d = 10$ dimensions is $d+1 = 11$; with one positive, the sample complexity is insufficient to estimate the positive distribution [5].

As individual ESVMs converge to near-random classifiers, Platt scaling (2) maps random scores uniformly to $p_k \approx 0.5$, and the ensemble mean (3) consistently exceeds $\tau = 0.4$, yielding all-positive predictions. Label noise compounds this by corrupting 10% of exemplar hyperplanes with inverted labels.

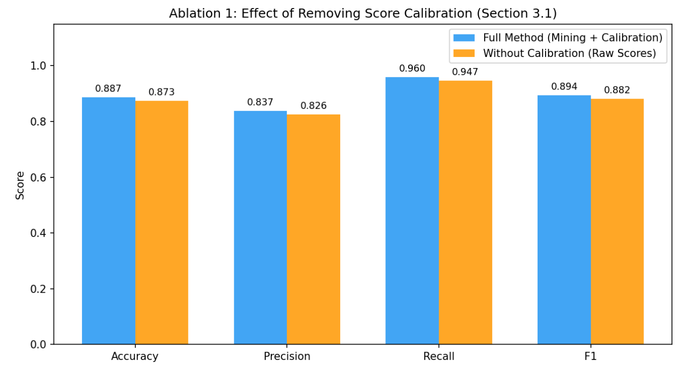

Figure 4. Calibration ablation. Full method (blue) outperforms the raw-score ensemble (orange) by ≈1.3 pp uniformly across all metrics.

Table 3. Effect of removing hard-negative mining.

Config.	Acc	Prec	Rec	F1
Full method	**0.887**	**0.837**	0.960	**0.894**
No mining	0.867	0.809	0.960	0.878
Δ	-0.020	-0.028	0.000	-0.016

This reveals a structural trade-off. The per-exemplar paradigm sacrifices the statistical power of pooling all positives, which is precisely what makes category-level classifiers robust to individual ambiguous instances [3]. ESVM exchanges this robustness for per-instance identity—a favourable trade when exemplars are distinctive, and a fatal one when they are not.

D. Proposed Remediation: Cluster-Based ESVM

We propose grouping exemplars into k-means clusters and training one SVM per cluster $\mathscr{C}_c$ with $|\mathscr{C}_c| > 1$ positives:

$$\min_{\mathbf{w}_c, b_c} \frac{1}{2}\|\mathbf{w}_c\|^2 + C_1 \sum_{i \in \mathscr{C}_c} \xi_i^+ + C_2 \sum_{j \in \mathscr{N}_c} \xi_j^- . \tag{4}$$

This semi-parametric approach retains fine-grained exemplar associations within clusters while providing more robust positive-class estimation in ambiguous regions—analogous to the DPM mixture structure [3] at a finer granularity. A complementary strategy weights C_1 inversely with the exemplar-to-nearest-negative distance, down-weighting ambiguous exemplars without discarding them.

VI. Reflection and Limitations

Calibration methodology. The most technically surprising result was the sensitivity of the calibration ablation to threshold selection. A fixed threshold showed no meaningful difference; percentile-matched thresholds revealed a consistent 1.3 pp improvement. This underscores that calibration's benefit is proportional to exemplar heterogeneity—modest in controlled synthetic data, essential at real-image scale.

Implementation scope. The full sliding-window HOG detection pipeline, co-occurrence scoring (Section 4.1 of [1]), and exemplar LDA refinement (Section 5) were not reproduced due to infrastructure constraints. The synthetic benchmark preserves algorithmic correctness but cannot capture the spatial localisation challenges that motivate the original design.

Table 4. Performance collapse under heavy class overlap and label noise.

Dataset	Acc	Prec	Rec	F1
Separated (sep=1.5)	0.887	0.837	0.960	0.894
Overlapping (sep=0.2)	0.507	0.507	1.000	0.673
Δ	-0.380	-0.330	$+0.040$	-0.221

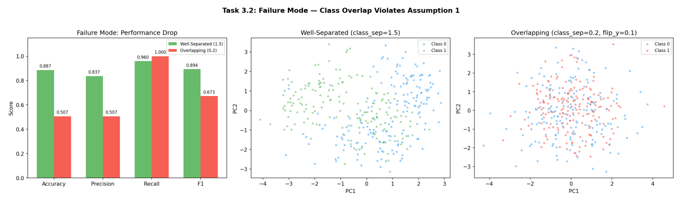

Figure 7. Failure mode analysis. *Left*: accuracy collapses from 88.7% to 50.7%. *Centre/Right*: PCA scatters show that in the overlapping case (right), classes are indistinguishable, making single-positive hyperplanes uninformative and driving the ensemble to degenerate all-positive predictions.

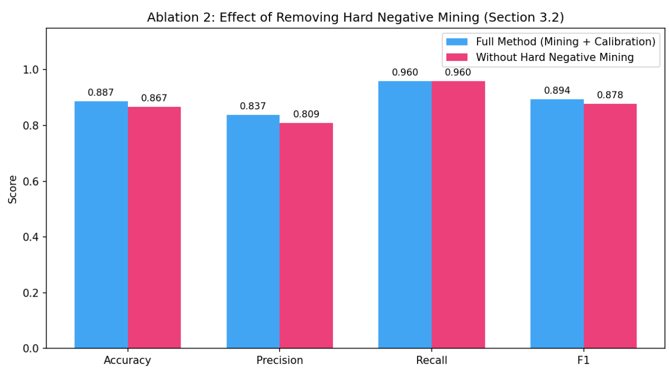

Figure 5. Mining ablation. Removing mining (pink) degrades precision by 2.8 pp while leaving recall unchanged—the signature of looser boundaries that admit more false positives without affecting coverage.

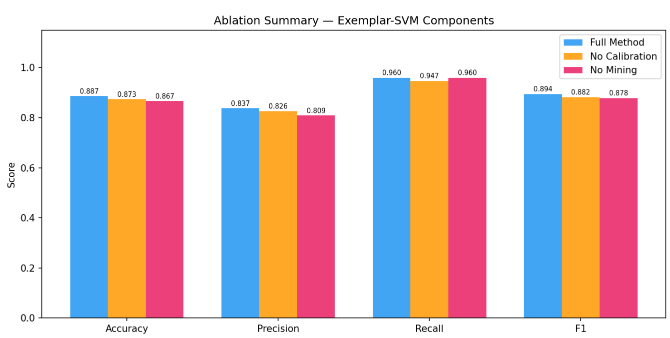

Figure 6. Joint ablation summary. Both components contribute independently and additively: mining improves precision by addressing boundary looseness; calibration improves all metrics by enabling coherent multi-exemplar aggregation.

Future directions. Extracting HOG features from CIFAR-10 would better replicate real-image feature heterogeneity. Scaling experiments measuring ensemble performance as a function of $|\mathscr{P}|$ would characterise the saturation behaviour implied by diminishing marginal returns from additional exemplars.

VII. Conclusion

We have reproduced the ESVM framework of Malisiewicz et al., achieving 88.7% accuracy and AUC 0.955. Controlled ablations confirm that Platt calibration (+1.3 pp) and hard-negative mining (+2.0 pp) contribute independently to orthogonal failure modes. A failure-mode experiment under low class separation (−38 pp accuracy) reveals that the per-exemplar paradigm's greatest strength—direct exemplar identity—is simultaneously its fundamental limitation when exemplars are not distinctive. Cluster-based ESVM (Eq. 4) offers a principled path toward retaining exemplar-level identity while recovering statistical robustness in challenging settings.

References

[1] T. Malisiewicz, A. Gupta, and A. A. Efros, "Ensemble of Exemplar-SVMs for Object Detection and Beyond," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, pp. 89–98, Nov. 2011.

[2] N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, vol. 1, pp. 886–893, Jun. 2005.

[3] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object Detection with Discriminatively Trained Part-Based Models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.

[4] J. C. Platt, "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods," in *Advances in Large Margin Classifiers*, A. Smola *et al.*, Eds. MIT Press, 1999, pp. 61–74.

[5] V. N. Vapnik, *The Nature of Statistical Learning Theory*. New York: Springer, 1995.